Riyadh Metro Data Engineering Platform

Project Architecture

Data Sources

- **RCRC Open Data** (Metro Stations & Lines)
- **API** (External Data)
- **SQL Database** (Operational Data)
- **CSV / Excel Files** (Additional Data)
- **Smart Metro Simulator** (Real-Time Streaming)

Batch Processing – Lakehouse Architecture

Data Pipeline

- Ingest
- Validate
- Transform
- Load

Lakehouse (Riyadh_Metro_Lakehouse)

Bronze Layer	Silver Layer	Gold Layer
bronze_events	silver_metro_events	Business Ready Data (Analytical)
bronze_metro_lines	silver_metro_lines	
bronze_metro_stations	silver_metro_stations	
bronze_metro_trips	silver_metro_trips	
bronze_weather		
Raw Data (Ingested)	Cleaned & Validated (Transformed)	

Data Warehouse (Riyadh_Metro_Warehouse)

- dim_lines
- dim_stations
- dim_events
- dim_weather
- fact_trips

Semantic Model (7 Tables)

- dim_lines
- dim_stations
- dim_events
- dim_weather
- fact_trips
- metro_line_paths
- UserStationAccess

Power BI Interactive Dashboard

- KPIs & Charts
- Filters
- Metro Map
- Line Paths
- Station Info
- Operational Data

Real-Time Processing – Streaming Architecture

Smart Metro Simulator (Real-Time Events) → **Eventstream** (Ingest & Process) → **Eventhouse** (Real-Time Analytics) → **metro_realtime_events** (Streaming Data)

Real-Time Data

Microsoft Fabric

Integrated Data Engineering Platform

OneLake · Lakehouse · Data Pipelines · Notebooks · Spark / PySpark · SQL · Data Warehouse · Semantic Model · Power BI · Eventstream · Eventhouse · KQL

From Multiple Sources → To Real-Time Insights